

Predicting Patient Medication Adherence

Using Machine Learning Classification Models

BUA 611 — Business Analytics

St. John's University

Professor Dr. Yanni Ping

Final Group Project Report

Shalini Chhetri, Anjali Rana, Carolina Zamudio

Table of Contents

1. Introduction	3
2. Data Description	3
3. Data Overview	4
4. Data Cleaning and Exploratory Analysis	4
4.1 Descriptive Summary of Key Metrics	4
4.2 Class Balance & Target Variable	4
4.3 Patient Demographics by Adherence Status	5
4.4 Medication Type Analysis	5
4.5 Healthcare Access & Insurance Distribution	5
4.6 Key Clinical Factors by Adherence Status	6
4.7 Correlation Analysis	6
5. Data Cleansing, Pre-Processing & Transformation	7
5.1 Encoding Categorical Variables	7
5.2 Feature Engineering	7
5.3 Standardization	7
5.4 Train/Test Split	8
6. Technology and Analysis	8
6.1 Machine Learning Approach	8
6.2 Classifier Parade & Model Selection	8
6.3 Decision Tree Classifier	9
6.3.1 Performance Evaluation	10
6.3.2 Classification Report Summary	10
6.4 Logistic Regression	10
6.4.1 Performance Evaluation	11
6.4.2 Classification Report Summary	12
7. Model Performance Comparison	12
8. Technical Challenges	13
9. Business Insights	14
10. Recommendations	14
11. Conclusion	15
12. References	16

1. Introduction

Medication adherence, the degree to which patients follow prescribed medication regimens, is a critical determinant of healthcare outcomes. Poor adherence leads to disease progression, increased hospitalizations, and higher healthcare costs. According to the World Health Organization (WHO, 2003), approximately 50% of patients with chronic conditions fail to adhere to their prescribed treatment plans.

This project analyzes patient-level data to build machine learning classification models that predict whether patients will adhere to their medication regimens. By identifying key predictors of non-adherence early, healthcare providers can design targeted interventions, allocate resources efficiently, and ultimately improve patient outcomes.

Using a dataset of 5,000 patient records sourced from Kaggle, we apply two classification models, the Decision Tree Classifier (DTC) and the Logistic Regression. Our analysis covers the full machine learning pipeline including data cleaning, feature engineering, cross-validation, and comprehensive model evaluation.

2. Data Description

The dataset was obtained from Kaggle and contains 5,000 observations and 14 variables, capturing a rich set of demographic, clinical, and socioeconomic features. The target variable is Adherence (binary: 1 = adherent, 0 = non-adherent).

Dataset link: <https://www.kaggle.com/datasets/rabieelkharoua/predicting-medication-adherence-dataset>

Variable Descriptions

Variable	Type	Description
Age	Numeric	Patient age in years (18–90)
Gender	Categorical	Male, Female, Other
Medication_Type	Categorical	Type of medication: Type A, Type B, Type C
Dosage_mg	Numeric	Prescribed dosage in milligrams (50–423mg)
Previous_Adherence	Binary	Was the patient previously adherent (1=Yes, 0=No)
Education_Level	Categorical	Highest education: High School, Graduate, Postgraduate
Income	Numeric	Annual household income (USD)
Social_Support_Level	Categorical	Level of social support: Low, Medium, High
Condition_Severity	Categorical	Disease severity: Mild, Moderate, Severe
Comorbidities_Count	Numeric	Number of comorbid conditions (0–6)
Healthcare_Access	Categorical	Access quality: Poor, Average, Good

Mental_Health_Status	Categorical	Mental health: Poor, Moderate, Good
Insurance_Coverage	Binary	Has insurance: 1= Yes, 0=No
Adherence	Binary	TARGET: 1=Adherent, 0=Non-Adherent

3. Data Overview

The dataset used for this project was sourced from Kaggle and covers a broad range of demographic, clinical, and socioeconomic characteristics. The dataset has no missing values or duplicate records, providing a reliable, clean foundation for machine learning analysis. The target variable is Adherence, a binary variable indicating whether the patient adhered to their medication regimen (1) or did not (0).

This dataset enables us to answer key clinical and business questions, including:

- Which patient characteristics most strongly predict medication adherence?
- Can we reliably flag at-risk non-adherent patients before adherence lapses?
- How do income, mental health, and social support influence adherence outcomes?

The overall goal is to leverage machine learning to generate actionable insights that improve patient adherence, inform targeted interventions, and support more effective clinical and population health strategies.

4. Data Cleaning and Exploratory Analysis

4.1 Descriptive Summary of Key Metrics

As part of the data cleaning and exploratory analysis process, we conducted a statistical summary of key numerical variables to better understand patient behavior and clinical trends. The dataset consists of 5,000 complete observations with no missing values and no duplicate records, ensuring strong reliability for analysis.

Age: The average patient age is approximately 50 years with a standard deviation of 14.6, ranging from 18 to 90. This wide range reflects a diverse patient population spanning young adults to elderly patients, which is clinically meaningful as adherence patterns often vary with age.

Dosage: The average prescribed dosage is 200.9mg with a standard deviation of 50.6, with a minimum of 50mg and a maximum of 423mg. The high variability in dosage suggests patients are on significantly different treatment intensities, which may influence their ability to maintain consistent adherence.

Income: Annual household income averages approximately \$501,000, indicating high variability across the patient population. This spread suggests the dataset captures both lower-income and higher-income patients, making income a potentially important socioeconomic predictor of adherence.

Comorbidities Count: Patients have an average of 2.0 comorbid conditions ranging from 0 to 6, indicating that most patients are managing multiple health conditions simultaneously. Higher

comorbidity burden is clinically associated with more complex medication regimens, which can negatively impact adherence.

Previous_Adherence: With a mean of approximately 0.70, roughly 70% of patients in the dataset had previously adhered to their medication, suggesting a baseline of moderate-to-good adherence history in this population.

Insurance Coverage: Approximately 80% of patients have insurance coverage, indicating that the majority of the sample has some form of financial protection for healthcare costs. The remaining 20% who are uninsured may face greater financial barriers to consistent medication adherence.

4.2 Class Balance & Target Variable

As part of our exploratory analysis, we examined the distribution of our target variable, Adherence, to assess whether the dataset is balanced between the two classes. Understanding class balance is a critical first step before model training because an imbalanced dataset can cause machine learning models to be biased toward predicting the majority class more frequently, resulting in misleadingly high accuracy while performing poorly on the minority class.

As shown in Figure 1, out of 5,000 patients in the dataset, 2,714 (54.3%) were non-adherent, and 2,286 (45.7%) were adherent to their medication regimen. While the split is relatively close, it is not perfectly balanced, meaning the data includes slightly more non-adherent patients than adherent ones. This mild class imbalance was considered throughout our modeling process. Rather than modifying the data itself, we selected evaluation metrics that account for imbalance, specifically F1 Macro Score¹ and ROC AUC², which measure model performance equally across both classes regardless of their size. This approach is consistent with best practices outlined in our course textbook.

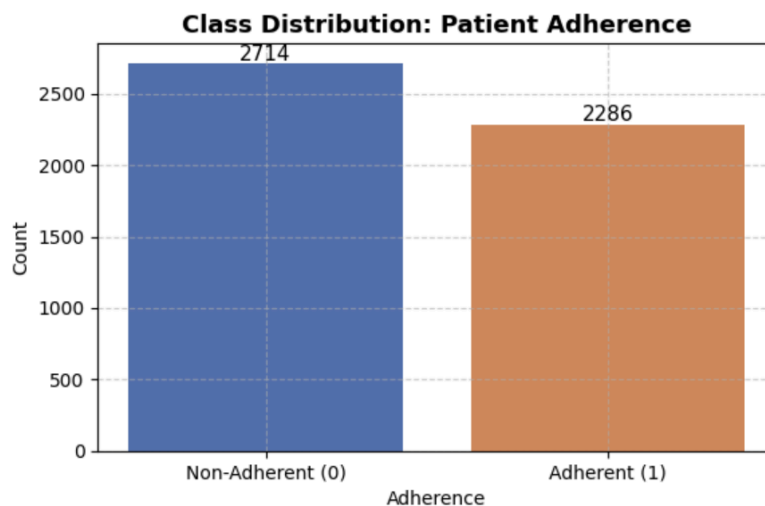


Figure 1: Class Distribution of Target Variable (Adherence)

4.3 Patient Demographics by Adherence Status

To better understand the patient population, we examined how demographic characteristics vary between adherent and non-adherent groups. As shown in Figure 2, Across gender categories, the distribution of adherent vs non-adherent patients appears relatively consistent between male and female patients, with the "Other" gender category showing slightly more variability due to its smaller sample size.

As illustrated in Figure 3, when examining education level, patients with postgraduate education show a slightly higher count of adherent individuals relative to non-adherent individuals, suggesting that health literacy and awareness may play a role in medication compliance. These demographic patterns provide useful context for understanding which patient subgroups may benefit most from targeted adherence interventions.

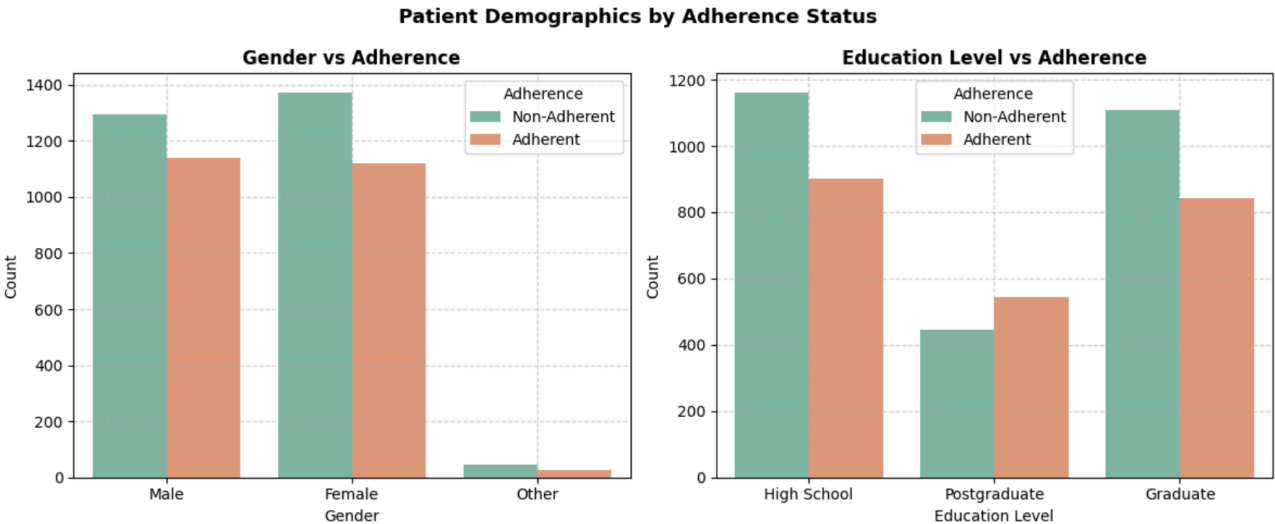


Figure 2: Gender vs (Adherence)

Figure 3: Education Level vs. (Adherence)

4.4 Medication Type Analysis

To assess whether the type of medication prescribed influences adherence behavior, we compared adherence patterns across the three medication categories present in the dataset: Type A, Type B, and Type C. As shown in Figure 4, the analysis reveals that the counts of adherent and non-adherent patients are relatively similar across all three medication types, suggesting that medication type alone is not a strong driver of non-adherence in this dataset. However, when combined with dosage complexity and condition severity, medication type may still contribute a meaningful signal to the predictive models. This finding supports the clinical understanding that adherence is a multifactorial behavior rather than being driven by any single variable.

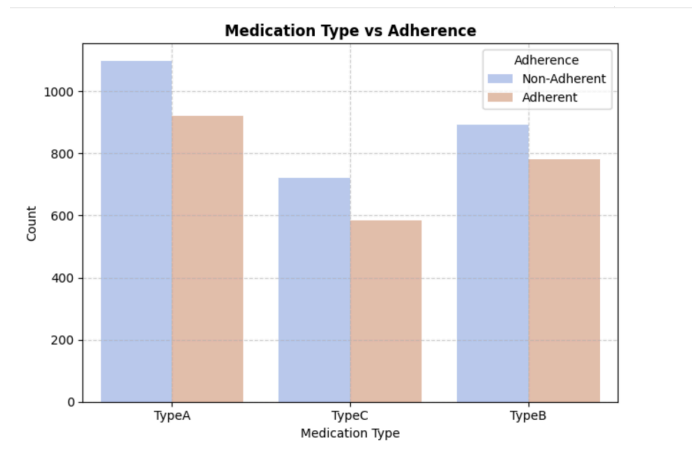


Figure 4: Medication Type vs. (Adherence)

4.5 Healthcare Access & Insurance Distribution

Access to healthcare and financial coverage are widely recognized as key barriers to medication adherence. As shown in Figure 5, patients with good healthcare access show a higher count of adherent patients compared to non-adherent patients within that group, reinforcing the importance of provider availability and patient-provider relationships in supporting consistent medication use.

Similarly, as illustrated in Figure 6, insured patients show a higher number of adherent individuals compared to uninsured patients, consistent with the broader clinical literature suggesting that financial barriers significantly affect a patient's ability to obtain and maintain prescriptions. Together, these two variables highlight the structural and socioeconomic dimensions of medication adherence that go beyond individual patient behavior.

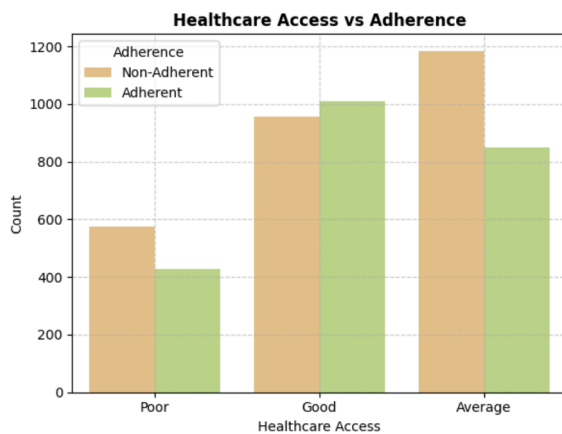


Figure 5: Healthcare Access vs Adherence

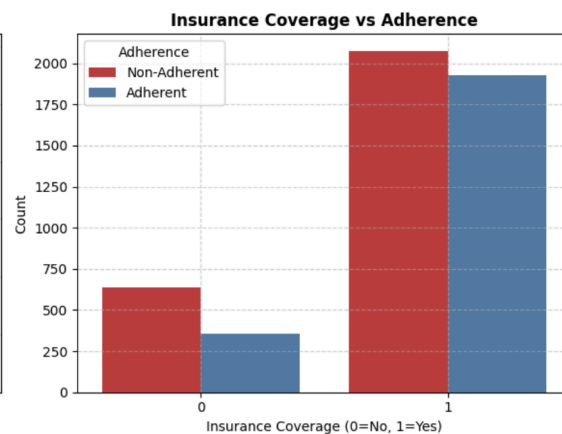


Figure 6: Insurance Coverage vs Adherence

4.6 Key Clinical Factors by Adherence Status

To deepen our understanding of adherence outcomes, we compared the average values of four key numerical variables, Age, Dosage_mg, Income, and Comorbidities_Count, across both patient groups. As shown in Figure 7, Income shows a notable difference between the two groups, with non-adherent patients having a higher average income compared to adherent patients. This suggests that income alone may not fully explain adherence behavior in this dataset and highlights the importance of considering multiple interacting factors. Dosage also shows a slight difference, with non-adherent patients on higher average dosages, suggesting that more complex regimens make adherence harder to maintain. Age and comorbidities remain relatively consistent across both groups, indicating they contribute some signal to the model but are not dominant predictors on their own. These findings are consistent with the feature importance results from our Decision Tree model discussed in Section 6.

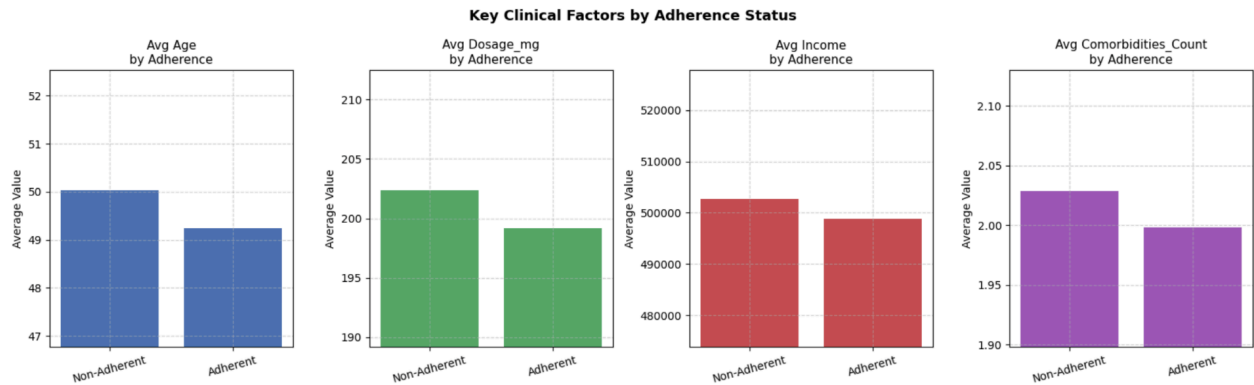


Figure 7: Key Clinical Factors by Adherence Status

4.7 Correlation Analysis

To examine the linear relationships between all variables in the dataset, we computed a correlation matrix and visualized it as a heatmap. Following the encoding step, the nominal variables Gender and Medication_Type are represented as binary One-Hot Encoded columns, Gender_1, Gender_2, MedType_1, and MedType_2, rather than single integer columns. Figure 8 shows that no single feature has a strong linear correlation with the target variable Adherence, with Previous_Adherence and Income showing the highest, though still modest, correlations. The individual One-Hot Encoded columns for Gender and Medication_Type each show near-zero correlation with Adherence, consistent with the EDA finding that these variables alone are not strong drivers of non-adherence. This confirms that medication adherence is a complex, multifactorial behavior that cannot be predicted by any single variable, which is precisely why machine learning models that can capture non-linear combinations of features are necessary for this task.

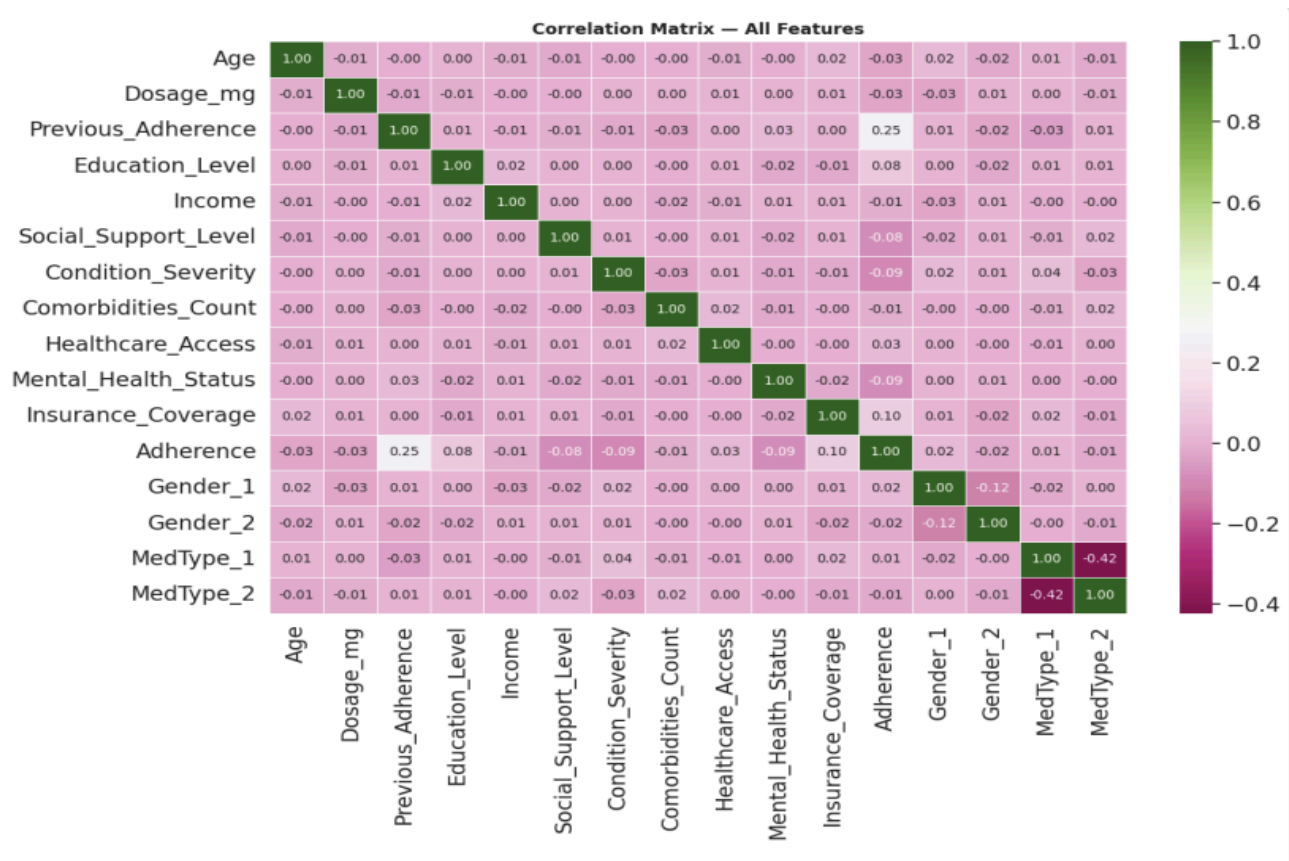


Figure 8: Correlation Matrix - All Features

5. Data Cleansing, Pre-Processing & Transformation

Before training any machine learning models, the dataset was cleaned, transformed, and prepared through a series of pre-processing steps to ensure it was suitable for analysis. This section summarizes each step taken to obtain the final training dataset.

5.1 Encoding Categorical Variables

Machine learning models require numerical inputs and cannot process raw text values directly. To address this, the seven categorical variables were encoded using two different strategies based on whether each variable is ordinal or nominal.

Five ordinal variables, Education_Level, Social_Support_Level, Condition_Severity, Healthcare_Access, and Mental_Health_Status, were encoded using scikit-learn's LabelEncoder, which assigns integers that preserve the natural ranking of categories. For example, Mental_Health_Status was encoded as Good=0, Moderate=1, and Poor=2.

Two nominal variables, Gender and Medication_Type were encoded using scikit-learn's OneHotEncoder with drop='first' to avoid multicollinearity. Since both variables contain three categories each, this produced two binary columns per variable after dropping the first category: Gender_1 and Gender_2 for

Gender, and MedType_1 and MedType_2 for Medication_Type. One-Hot Encoding was selected for these variables because they have no natural numeric ordering, and applying Label Encoding would incorrectly imply a hierarchy between categories such as Male, Female, and Other.

The remaining variables were already in numerical format and required no encoding. A copy of the original dataframe was preserved throughout this process to ensure that the EDA visualizations retained their readable text labels.

5.2 Feature Engineering

To enrich the dataset with additional contextual information, one new feature was created. Avg_Income_by_Education was calculated as the average income of all patients sharing the same education level group. This feature captures the broader socioeconomic context associated with each education level and provides the model with a more generalized financial signal beyond individual income alone. To prevent data leakage, this feature was computed exclusively from the training set after the train/test split, using a groupby mean on Education_Level. The resulting group means were then mapped to the test set, ensuring no test-set information influenced the engineered feature values.

5.3 Standardization

All numerical features, Age, Dosage_mg, Income, Comorbidities_Count, and Avg_Income_by_Education were standardized using scikit-learn's StandardScaler after the train/test split. The scaler was fit exclusively on the training set using fit_transform, and the learned mean and standard deviation were then applied to the test set using transform only. This ensures that no information from the test set influenced the scaling parameters, preventing data leakage. Standardization is particularly important for Logistic Regression, where features on different scales can cause unstable coefficient estimates and slow convergence.

5.4 Train/Test Split

The processed dataset was split into a training set of 67% and a test set of 33% using scikit-learn's train_test_split function with a fixed random state of 21 to ensure reproducibility. The training set containing 3,350 observations was used exclusively for model training and cross-validation. The test set containing 1,650 observations was held out entirely and used only for final model evaluation, providing an unbiased assessment of how each model performs on completely unseen data. Feature engineering and standardization were both applied after this split to ensure the held-out test set remained completely unseen during all preprocessing steps, with scaler parameters and group means derived solely from the training data.

6. Technology and Analysis

6.1 Machine Learning Approach

This project applies a supervised machine learning classification approach to predict whether a patient will adhere to their prescribed medication regimen. Since the target variable Adherence is binary, either adherent (1) or non-adherent (0), classification models are the most appropriate method for this task.

Following the methods covered in Chapter 8 of the course textbook, we selected two classification models for detailed evaluation: the Decision Tree Classifier (DTC) and Logistic Regression. These two models were chosen based on their performance in the classifier parade, their alignment with course material, and their complementary strengths. The Decision Tree offers high interpretability through visual decision rules and feature importances, while Logistic Regression provides a stable probabilistic baseline with interpretable coefficients that directly indicate the direction and strength of each feature's influence on adherence.

All models were evaluated using F1 Macro Score and ROC AUC as the primary metrics, as both account for the mild class imbalance in the dataset and measure performance equally across both classes.

6.2 Classifier Parade & Model Selection

To systematically identify the best classification models for this dataset, we conducted a classifier parade, evaluating 10 different algorithms using 5-fold cross-validation with F1 Macro Score as the evaluation metric. The models tested included two Logistic Regression variants, Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Gaussian Naive Bayes, Support Vector Machines (linear and kernel-based), Decision Tree Classifier, and K-Nearest Neighbors (5NN and 10NN).

F1 Macro Score was selected as the evaluation metric rather than plain accuracy because it equally weights performance on both the adherent and non-adherent classes, making it more informative and reliable for a mildly imbalanced dataset. As shown in Figure 9, the cross-validation results across all five folds demonstrate that the Decision Tree and Logistic Regression were selected for further analysis due to their strong and consistent performance across cross-validation, as well as their interpretability and alignment with course objectives.

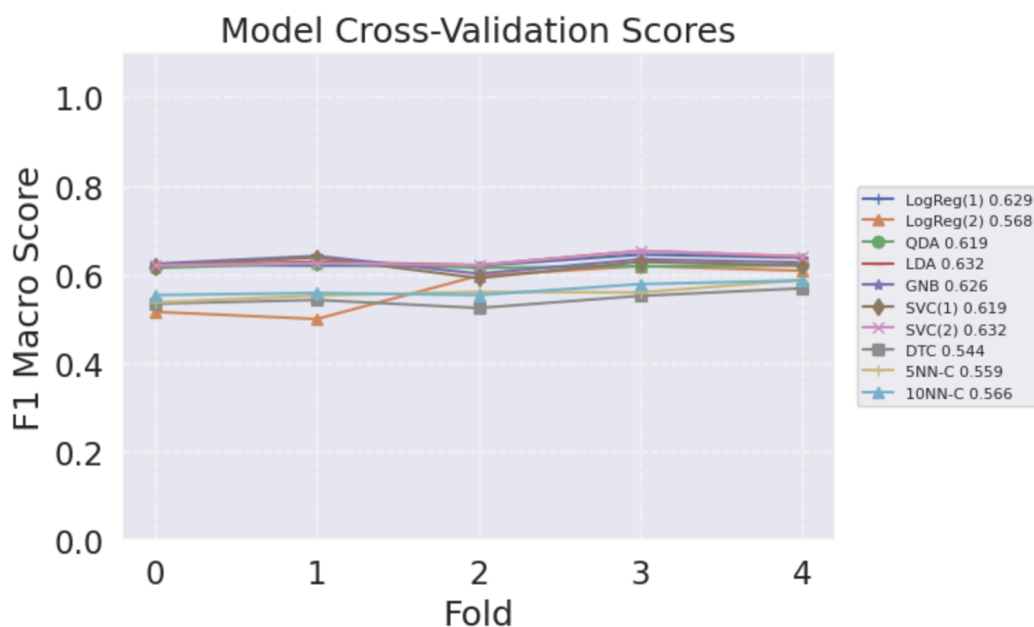


Figure 9: Model Cross Validation Scores

6.3 Decision Tree Classifier

The model was first used without any depth restriction to examine feature importances, identifying which variables contribute most to predicting adherence. Following One-Hot Encoding, Gender and Medication_Type are represented as individual binary columns, Gender_1, Gender_2, MedType_1, and MedType_2, each with their own importance score. As shown in Figure 10, Dosage_mg, Income, and Age remained the top three predictors, consistent with our earlier EDA findings. The individual One-Hot Encoded columns for Gender and Medication_Type each show relatively low importance scores, confirming that these variables are weaker predictors of adherence compared to clinical and socioeconomic factors.

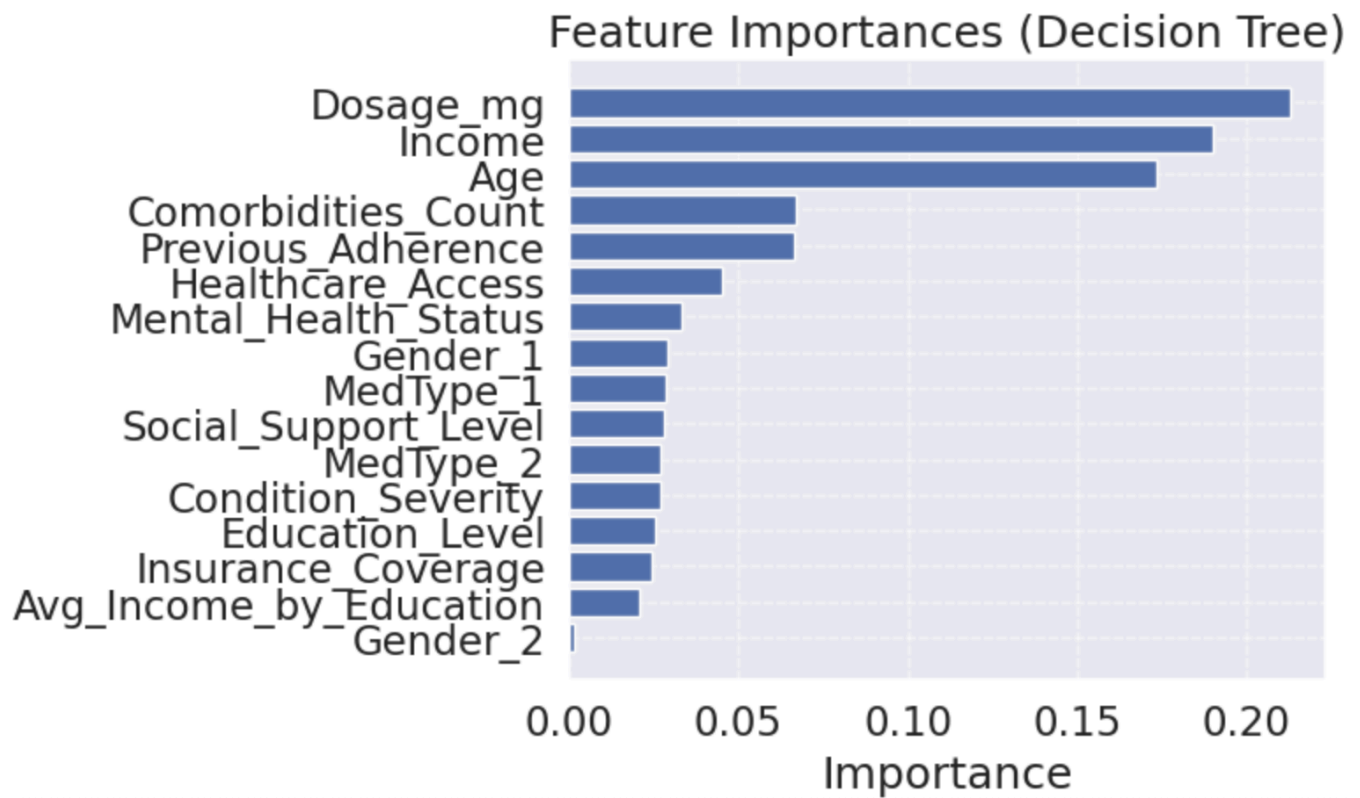


Figure 10: Feature Importances (Decision Tree)

The model was then tuned with a maximum depth of 3 to prevent overfitting. As shown in the decision boundary plots in Figure 11, an unrestricted tree creates highly irregular boundaries that indicate memorization of the training data rather than learning generalizable patterns. At `max_depth=1` the model underfits severely. A depth of 3 was selected as a practical balance between model complexity and generalization. While the decision boundary plots are based on a two-dimensional projection of the data, they provide a useful visual illustration of how model complexity changes with tree depth. Cross-validation results further support this choice by demonstrating stable performance without overfitting.

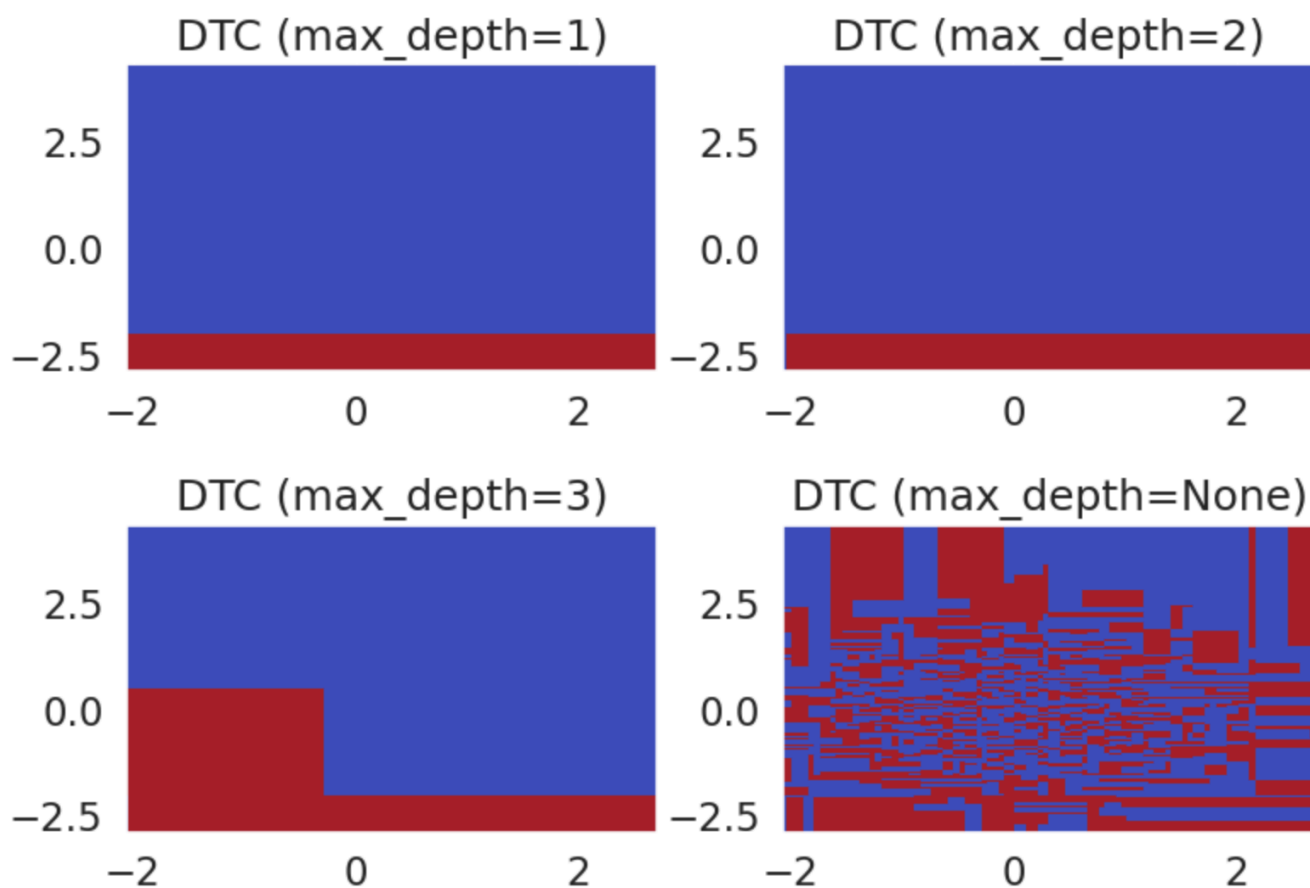


Figure 11: DTC Maximum Depth

The tree structure visualization in Figure 12 shows that the first split is based on `Previous_Adherence`, indicating it was the best early split for separating adherence outcomes in the tuned tree. At the same time, the unrestricted tree's feature importance ranked `Dosage_mg` and `Income` among the strongest overall predictors.

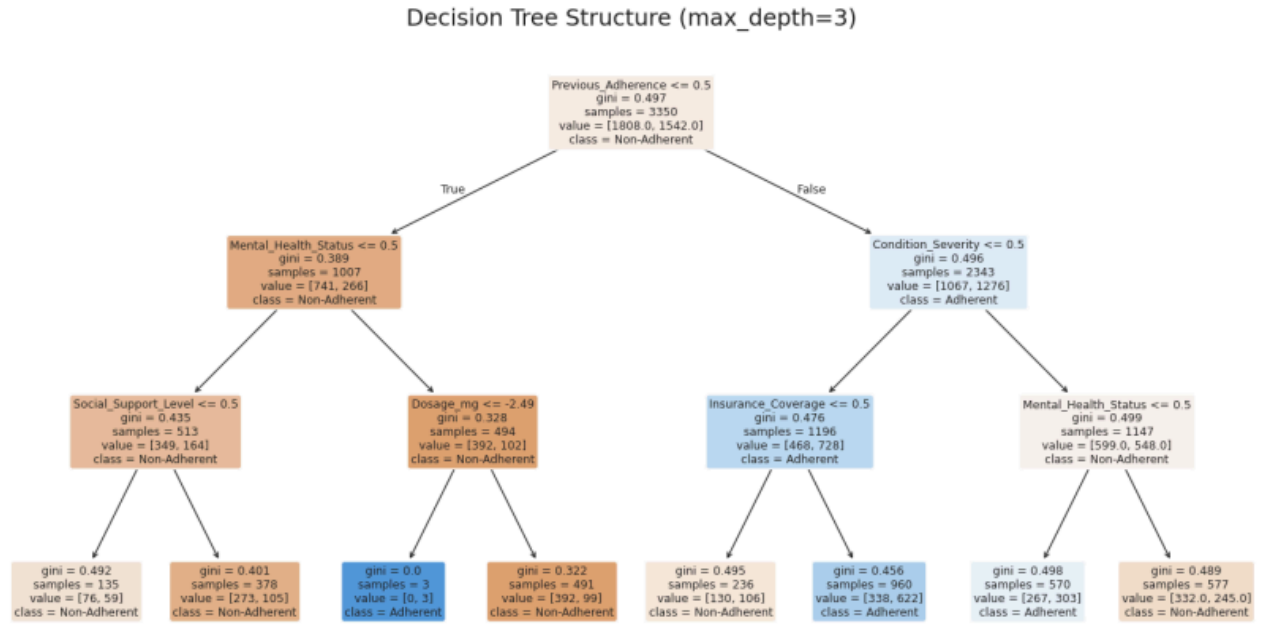


Figure 12: Decision Tree Structure

6.3.1 Performance Evaluation

To evaluate the Decision Tree Classifier, we applied 5-fold cross-validation on the training set using F1 Macro Score as the primary metric, ensuring that performance on both adherent and non-adherent patients was equally considered. The model achieved a mean F1 Macro Score of 0.611 across the five folds, demonstrating consistent and stable performance.

On the held-out test set, the model achieved an accuracy of 61.8%. While accuracy is reported for interpretability, F1 Macro Score was used as the primary evaluation metric due to the mild class imbalance. As shown in Figure 13, the model correctly predicted 571 non-adherent patients and 448 adherent patients. It also misclassified 335 non-adherent patients as adherent and 296 adherent patients as non-adherent. This indicates the model performs slightly better at identifying non-adherent patients than adherent ones.

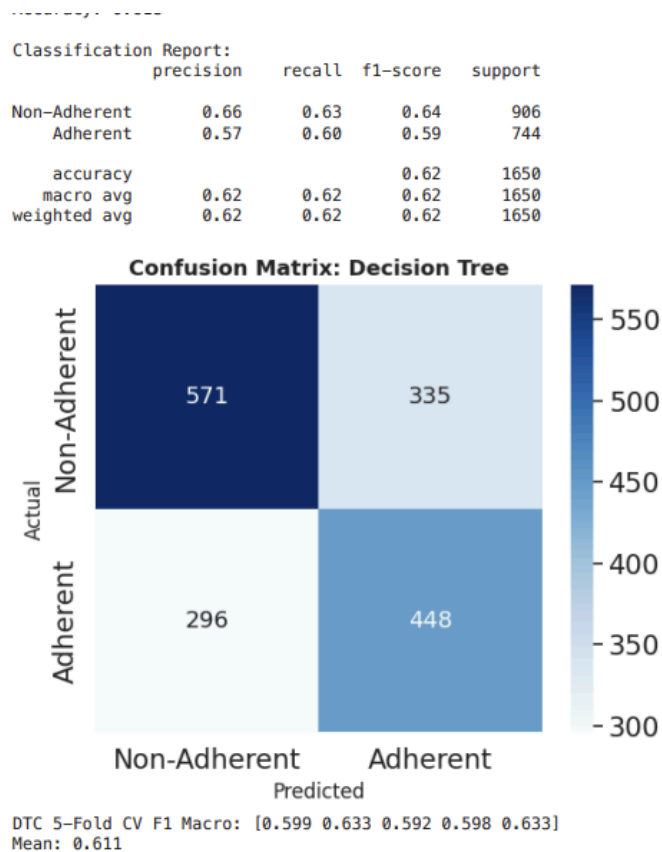


Figure 13: Decision Tree Confusion Matrix

6.3.2 Classification Report Summary

The classification report confirms these findings. For the non-adherent class (0), the model achieved a precision of 0.66 and recall of 0.63, resulting in an F1 score of 0.64. For the adherent class (1), precision was 0.57 and recall was 0.60, giving an F1 score of 0.59. The macro average F1 score of 0.62 reflects balanced but moderate performance across both classes. While the overall accuracy is modest, the Decision Tree provides significant value through its interpretability, clearly identifying Previous_Adherence, Dosage_mg, and Income as the primary clinical drivers of non-adherence.

6.4 Logistic Regression

Logistic Regression was selected as our second Chapter 8 model due to its consistent performance in the classifier parade and its strong interpretability through feature coefficients. Unlike the Decision Tree which partitions the feature space into regions, Logistic Regression estimates the probability of adherence using a linear combination of features transformed through the sigmoid function, producing probabilities between 0 and 1.

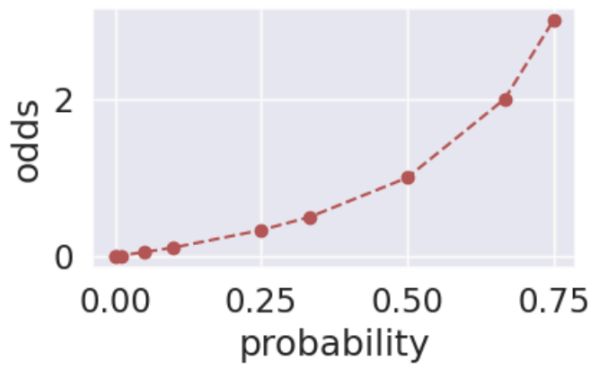


Figure 14: Logistic Regression vs. Probability

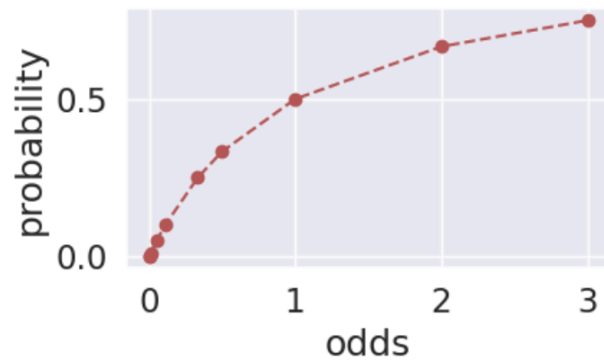


Figure 15: Logistic Regression vs. Odds

Figures 14 and 15 illustrate the mathematical foundation of Logistic Regression. The model estimates the probability of adherence by transforming a linear combination of features into probabilities using the sigmoid function. This transformation converts log-odds into values between 0 and 1, allowing the model to output interpretable probabilities for classification.

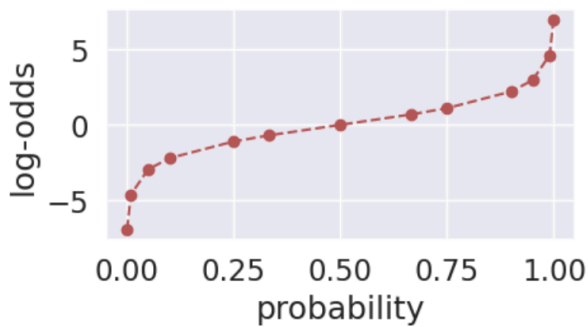


Figure 16: Log odds vs. Probability

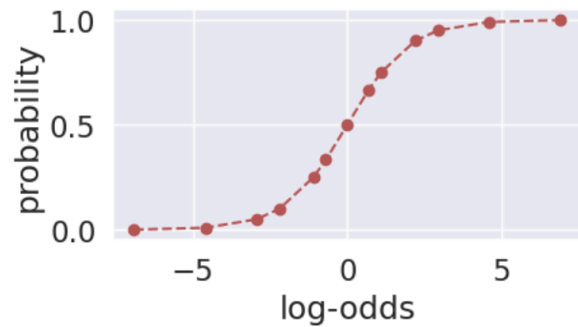
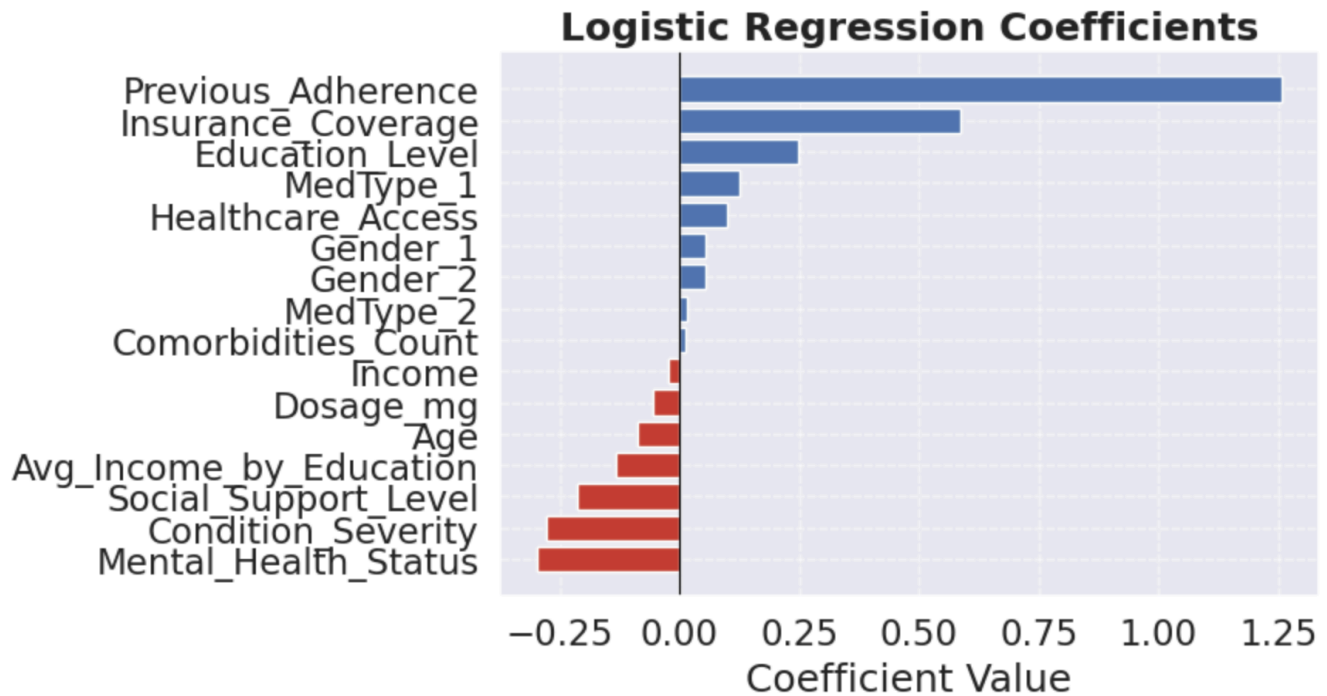


Figure 17: Log-Odds to Probability (Sigmoid Curve).



Figures 16 and 17 compare the decision boundaries produced by two optimization methods, SAGA and SGD. As shown in the 2D visualization in Figure 19, SAGA produced a more useful decision boundary than SGD, so SAGA was selected for the final Logistic Regression model. The SAGA solver, while also predominantly predicting adherence, produces a partial decision boundary in the lower-left region, demonstrating at least some ability to distinguish between classes. Based on this comparison, SAGA was selected for the final Logistic Regression model as the more reliable optimization method.

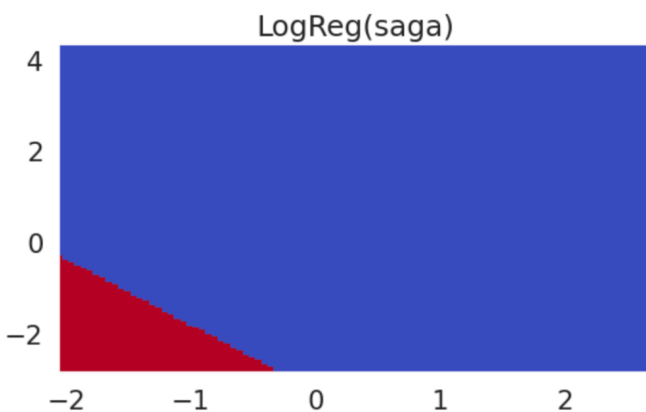


Figure 18: Log odds vs. Probability

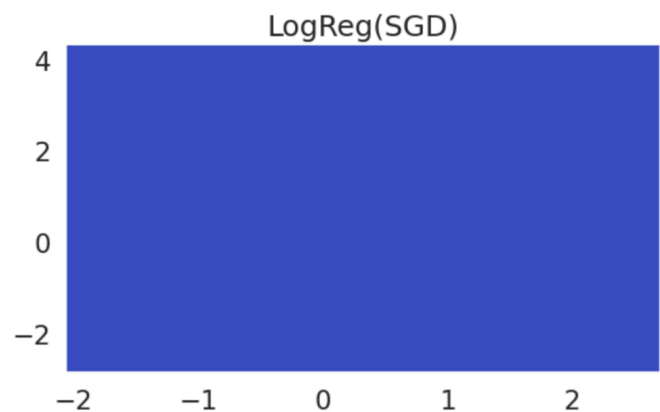


Figure 19: Logistic Regression — Log-Odds to Probability (Sigmoid Curve).

As shown in Figure 18 and 19, the coefficient plot reveals that Previous_Adherence, Insurance_Coverage, and Education_Level contribute most positively toward predicting adherence, while poor Mental_Health_Status and poor Condition_Severity are most strongly associated with non-adherence. Following One-Hot Encoding, Gender and Medication_Type are represented as binary columns, Gender_1, Gender_2, MedType_1, and MedType_2, each with individual coefficients reflecting the effect of belonging to that specific category relative to the dropped reference category. The

magnitudes of these OHE coefficients are modest, confirming that Gender and Medication_Type are weaker predictors of adherence compared to clinical and socioeconomic factors. These findings are directly interpretable and actionable for clinical decision-making.

Figure 20: Logistic Regression Coefficients

6.4.1 Performance Evaluation

To evaluate the Logistic Regression model, we applied 5-fold cross-validation on the training set using F1 Macro Score, achieving a mean score of 0.629, slightly higher than the Decision Tree. On the held-out test set, the model achieved an accuracy of 62.7%. As shown in Figure 21, the model correctly predicted **581** non-adherent patients and **454** adherent patients, with **325** false positives and **290** false negatives. Compared to the Decision Tree, Logistic Regression shows more balanced performance across both classes with a notably higher ROC AUC of **0.676** versus **0.649** for the Decision Tree.

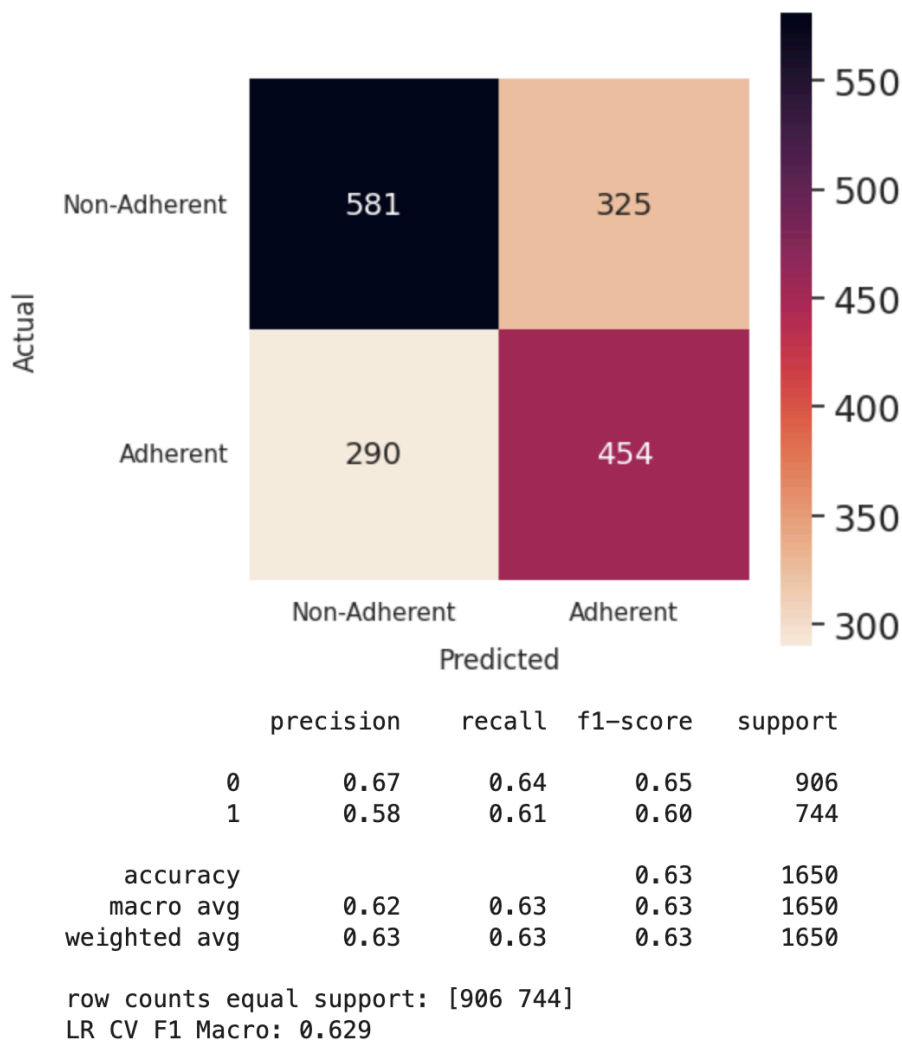


Figure 21: Logistic Regression Confusion Matrix

6.4.2 Classification Report Summary

For the non-adherent class (0), the model achieved a precision of 0.67 and recall of 0.64, giving an F1 score of 0.66. For the adherent class (1), precision was 0.59 and recall was 0.62, giving an F1 score of 0.60. The macro average F1 score of 0.63 and ROC AUC of 0.675 confirm that Logistic Regression is the stronger and more reliable model for predicting patient medication adherence in this dataset.

7. Model Performance Comparison

The table and chart below summarize the cross-validated performance metrics for both models. As shown in Figure 22 and 23, Logistic Regression outperforms the Decision Tree across all four metrics, most notably in ROC AUC (+0.027), which measures the model's ability to discriminate between adherent and non-adherent patients across all possible decision thresholds.



Figure 22: Model Performance Comparison

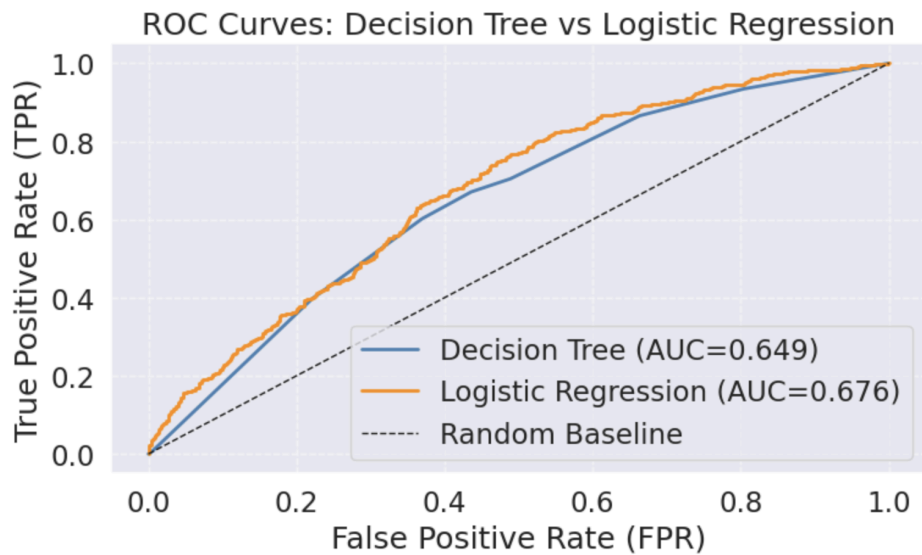


Figure 23: ROC Curve Comparison

Both models perform above the random baseline ($AUC > 0.5$). Logistic Regression is the preferred model for deployment due to its higher AUC, more balanced recall across classes, and interpretable coefficients that provide clinical insight into which factors drive non-adherence.

8. Technical Challenges

Several technical challenges arose during this project:

Class Imbalance: With 54.3% non-adherent vs 45.7% adherent patients, models without correction could favor predicting non-adherence. We addressed this by selecting evaluation metrics that account for class imbalance, specifically F1 Macro Score and ROC AUC, which measure performance equally across both classes regardless of size. This is consistent with the approach recommended in the course textbook.

Categorical Encoding Strategy: Selecting the appropriate encoding method for categorical variables required careful consideration of each variable's measurement scale. Nominal variables such as Gender and Medication_Type have no natural numeric ordering, meaning Label Encoding would incorrectly imply a hierarchy between categories and potentially mislead the Logistic Regression model. To address this, One-Hot Encoding with `drop='first'` was applied to these two variables following the approach outlined in the Chapter 10 course notebook. This expanded the feature space by adding four binary columns in place of the original two, which required ensuring all downstream steps including the correlation heatmap, feature importance chart, and coefficient plot correctly reflected the new column names.

Overfitting in Decision Tree: Unrestricted tree depth resulted in perfect training accuracy but significantly degraded test performance, a classic overfitting symptom. Setting `max_depth=3` after visual inspection of decision boundary complexity resolved this issue.

Logistic Regression Convergence: With high-dimensional features and varied scales, we encountered convergence warnings using default settings. Applying StandardScaler before training and using the SAGA solver with max_iter=1000 resolved the convergence issue reliably.

Feature Engineering: Creating the Avg_Income_by_Education feature required careful group-merge operations using pandas, with attention to ensuring no data leakage from the test set into the feature calculation.

Visualization Scaling: With 5,000 data points, maintaining visual clarity was a key challenge. Plots were thoughtfully formatted and structured to ensure interpretability without compromising the use of the full dataset. All machine learning models were trained and evaluated using the complete dataset of 5,000 observations.

9. Business Insights

The analysis yields several actionable insights for healthcare providers and policymakers:

Previous_Adherence was the root split in the tuned tree and the strongest positive Logistic Regression Coefficient while the unrestricted tree's feature importance ranked Dosage_mg and Income highly. This enables healthcare systems to proactively flag patients who have previously struggled with adherence before prescriptions are renewed.

Income data shows that non-adherent patients had comparable or higher incomes than adherent ones. Financial assistance programs alone may be misallocated — healthcare access quality, social support, and mental health are the more impactful levers for intervention.

Mental health status is strongly associated with adherence. Patients with poor mental health are significantly less likely to adhere, supporting the integration of behavioral health support into medication management programs.

Insurance coverage markedly improves adherence probability. Policy interventions expanding insurance access may improve population-level adherence more efficiently than targeted individual interventions.

Dosage complexity negatively impacts adherence: patients on higher dosages show lower adherence, suggesting that simplified dosing schedules or combination therapies may improve compliance.

10. Recommendations

Based on the machine learning findings, we recommend the following:

Deploy Logistic Regression as the Primary Screening Model: Its higher ROC AUC (0.675) and interpretable coefficients make it the most clinically useful model. Healthcare systems can integrate it into EHR systems to flag at-risk patients automatically at prescription time.

Income-Based Intervention Targeting: Income should not be used alone for targeting, interventions should prioritize healthcare access, insurance, mental health, and social support. Connecting low-income patients with prescription assistance resources at the point of prescription could yield measurable results.

Integrate Mental Health Screening: Routinely screen patients for mental health status at medication initiation. Patients identified with poor mental health should receive coordinated behavioral health referrals alongside their medication plans.

Simplify High-Dosage Regimens: Work with prescribers to simplify complex dosage regimens wherever clinically possible. Extended-release formulations and once-daily dosing options should be explored for patients flagged as high non-adherence risk.

Continuous Model Monitoring: Retrain models quarterly as new patient data becomes available. Periodic recalibration ensures the model adapts to evolving patient demographics and treatment patterns.

Overall, the model demonstrates that medication adherence is driven by a combination of behavioral, clinical, and socioeconomic factors rather than any single variable.

11. Conclusion

This project successfully demonstrated the practical application of supervised machine learning to predict patient medication adherence, a high-impact problem in healthcare management. Starting from a clean, well-structured dataset of 5,000 patients with 14 features, we conducted end-to-end analysis: exploratory data analysis, feature engineering, careful metric selection to address class imbalance, and comparative modeling using Decision Tree Classifier and Logistic Regression.

Among the two Chapter 8 classification models evaluated, Logistic Regression emerged as the stronger performer with an accuracy of 62.7%, F1 Macro of 0.63, and ROC AUC of 0.675. The Decision Tree Classifier, while slightly less accurate, offered superior interpretability through its feature importance rankings and tree visualization, providing complementary clinical insights.

Key predictors of medication adherence identified in this study include Previous_Adherence behavior, income, dosage complexity, insurance coverage, and mental health status. These findings directly translate into actionable business recommendations: financial assistance programs, mental health integration, insurance expansion, and predictive screening tools integrated into electronic health records.

Future work could explore more advanced ensemble methods such as Random Forest and Gradient Boosting, neural network approaches for non-linear interaction detection, and real-time deployment pipelines that score patients at prescription initiation to trigger immediate adherence support interventions.

12. References

El Kharoua, R. (2024). *Predicting medication adherence dataset* [Data set]. Kaggle. <https://www.kaggle.com/datasets/rabieelkharoua/predicting-medication-adherence-dataset>

World Health Organization. (2003). *Adherence to long-term therapies: Evidence for action*. World Health Organization. <https://www.paho.org/sites/default/files/WHO-Adherence-Long-Term-Therapies-Eng-2003.pdf>

Fenner, M. E. (2020). *Machine learning with Python for everyone*. Addison-Wesley Professional.

scikit-learn developers. (2024). *scikit-learn: Machine learning in Python*. <https://scikit-learn.org>